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$\Rightarrow \frac{a_2}{a_1} = \frac{C_2}{C_1} \Rightarrow \mu_2 - \mu_1 = RT\ln\left(\frac{C_2}{C_1}\right)$

However, $C_2 < C_1 \Rightarrow \mu_2 - \mu_1 < 0 \Rightarrow \mu_2 < \mu_1$

Therefore, the chemical potential of water in solution (μ_2) is lower than that of pure water (μ_1).

5 - OSMOLARITY OF A SOLUTION

For a solution containing a single solute, the osmolarity is defined as:

$$\omega = [1 + \alpha(\beta - 1)]C$$

- $\omega = \frac{n_{osmoles\ of\ soluty}}{V_{solution}}$: **osmolarity** of the solution. $[\omega] = osmol/l$
- $C = \frac{n_{moles\ of\ soluty}}{V_{solution}}$: **molarity** of the solution. $[C] = mol/l$
- β : **number of ions** produced by the dissociation of the solute, if it is an **electrolyte**.

  $\beta = 1$ if it is a **non-electrolyte** (e.g., urea and glucose) $\Rightarrow \omega = C$

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Examples:

$N_aCl \rightarrow N_a^+ + Cl^- \Rightarrow \beta = 2$

$N_{a_2}SO_4 \rightarrow 2N_a^+ + SO_4^{2-} \Rightarrow \beta = 3$

- α : dissociation coefficient of **Vant'Hoff**.

$$\alpha = \frac{Number\ of\ dissociated\ moles}{Initial\ number\ of\ moles} = \frac{n_{dissociated}}{n_{initial}}$$

$0 \leqslant \alpha \leqslant 1$

Examples:

- HCl, **strong** electrolyte: $HCl \rightarrow H^+ + Cl^-$
  number of dissociated moles = initial number of moles $\Rightarrow \alpha = 1$
- $HCOOH$, **weak** electrolyte : $HCOOH \rightarrow H^+ + HCOO^-$
  number of dissociated moles < initial number of moles $\Rightarrow \alpha < 1$
- Glucose, **non-electrolyte**.
  number of dissociated moles $= 0 \Rightarrow \alpha = 0$

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Remarks

- If the solution is composed of i solutes $(\alpha_i, \beta_i, C_i, \omega_i)$, the total osmolarity is given by:

$$\omega = \sum_i \omega_i = \sum_i [1 + \alpha_i(\beta_i - 1)]C_i$$

- **Osmolality** is used when C is expressed as **molality**.

$$\omega = [1 + \alpha(\beta - 1)]C$$

$\omega = \frac{n_{osmoles\ of\ soluty}}{m_{solvent}}$: **osmolality** of the solution. $[\omega] = osmol/kg$

$C = \frac{n_{moles\ of\ soluty}}{m_{solvent}}$: **molality** of the solution. $[C] = mol/kg$

Exercise 5:

In $1l$ of water, there are $5.85g$ of $NaCl$ and $18g$ of glucose $(180g/mol)$. Calculate the osmolarity of the solution.

Solutions:

$\omega = \omega_{NaCl} + \omega_{Glucose}$

$*$ *NaCl* :

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$$\omega_{NaCl} = [1 + \alpha(\beta - 1)]C_{NaCl}$$

$$\alpha = 1; \beta = 2 \Rightarrow \omega_{NaCl} = 2C_{NaCl} = 0.2osmol/l$$

$$C_{NaCl} = \frac{n}{V} = \frac{m/M}{V} = \frac{m}{MV} = \frac{5.85}{58.5\times1} = 0.1mol/l$$

$*$ *Glucose* :

$$\omega_{Glucose} = C_{Glucose} = 0.1osmol/l$$

$$C_{Glucose} = \frac{m}{MV} = \frac{18}{180\times1} = 0.1mol/l$$

$$\omega = 0.2 + 0.1 = 0.3osmol/l = 300mosmol/l$$

F- TYPES OF VESSELS AND MEMBRANES

- The 5 types of blood vessels: arteries, arterioles, capillaries, veins and venules.

Blood : $heart \xrightarrow{arteries} organs \xrightarrow{veins} heart$

Gases and nutrients : $blood \xleftrightarrow{capillaries} tissues$

